Prabhu Manikandan Viswanathan

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PROFESSIONAL SUMMARY

Cybersecurity graduate with hands-on experience in vulnerability assessments, penetration testing, threat intelligence, and cyber governance. Proven ability to work in fast-paced environments through academic, internship, and research roles. Demonstrated strengths in technical problem-solving, teamwork, and communication. Passionate about helping clients enhance their security posture and mitigate cyber risks through proactive defence strategies and compliance best practices.

SKILLS

Penetration Testing, Risk Assessment, Threat Intelligence, Cybersecurity Audits, ISO 27001, NIST, Nmap, Burp Suite, sqlmap, Metasploit, DirBuster, Linux, Windows, Network Security, Python, Bash, PowerShell, ISO 27001, NIST, CMMI, Log Analysis, SIEM (Splunk, Elastic Stack), Wireshark, Docker, AI/ML in Cyber Security.

EXPERIENCE

Research Assistant *September 2024 – March 2025*
University of Warwick *Coventry, United Kingdom*

ResAuto

- Developed vulnerability proof-of-concepts in CheriBSD for automotive embedded systems using Lua and embedded C, simulating memory corruption and privilege escalation attacks to assess compartmentalization effectiveness.
- Analysed exploit behaviours and system responses to craft mitigation strategies, contributing to the design of secure embedded architectures and enhancing detection capabilities for real-world malware targeting automotive platforms.

Cyber Observatory (The Alan Turing Institute)

- Conducted threat intelligence research for emerging IoC/Attack, exploits and vulnerabilities in critical infrastructure security, national security.
- Leveraged MISP and OpenCTI to analyse Indicators of Compromise (IOCs) and enhance incident detection & response.
- Generated threat trend reports and dashboards to support national security and strategic decision-making.

Cyber Security Intern *February 2023 - April 2023*
TIAA GBS India *Pune, India*

- Automated risk assessment processes, reducing supplier risk evaluation time from 2 weeks to 20 seconds, enhancing audit efficiency.
- Performed in-depth log analysis using Splunk to detect and investigate security incidents.
- Hands-on experience working in a Security Operations Center (SOC) environment, conducting real-time threat detection, analysis, and response across enterprise networks.
- Streamlined cybersecurity maturity assessments using the CMMI framework, enabling stakeholders and senior management to easily understand subsidiary cybersecurity posture.

Project Trainee – ISMS *July 2022 - August 2022*
Larsen & Toubro EAIC *Mumbai, India (Remote)*

- Conducted ISO 27001 internal audits, resolved major non-conformities, and ensured compliance with security controls and governance policies.
- Trained 100+ employees in cybersecurity best practices, enhancing audit preparedness and overall compliance awareness.

EDUCATION

Master of Science in Cyber Security Engineering *(Distinction)* *October 2023 - October 2024*
The University of Warwick (NCSC Certified) *United Kingdom*

Bachelor of Technology (Hons.) in Computer Science Engineering with specialisation in Cyber Security and Forensics *August 2019 - May 2023*
University of Petroleum and Energy Studies *Dehradun, India*

CERTIFICATIONS

- ISO 27001 Information Security Management Systems Lead Auditor – *BSI*
- CompTIA Security+ (*In Progress, Expected 08/2025*)

ACHIEVEMENTS

- **Black Hat USA 2024**: Awarded a student scholarship to attend the prestigious cybersecurity conference.
- **ISACA Digital Scholarship 2024**: Recipient of the international digital scholarship.

PUBLICATIONS

Implementing Anti-virus Evasion - Process Ghosting Attack and Prevention (IEEE)

A Novel Approach for an Automated Advanced MITM Attack on IoT Networks (Springer)

Cyber Forensics and Investigation on Smart Devices (e-book)

PROJECTS

Efficiency of XAI in Human-Machine Teaming for Cyber Security *(Master's Dissertation)* **(2024)**

- Developed the HxC Framework integrating Explainable AI (XAI) with Human-Machine Teaming (HMT) to enhance trust, transparency, and collaboration in risk detection and threat hunting.
- Proposed ethical and interpretable AI solutions, addressing scalability and adaptability challenges in risk and threat assessments, making AI more transparent and auditable.

Penetration Testing and Vulnerability Assessment (2024)

- Conducted all critical phases of a penetration test, including reconnaissance, scanning, vulnerability assessment, exploitation, and post-exploitation, using tools like Nmap, sqlmap, DirBuster, Metasploit, and Mimikatz.
- Identified and exploited vulnerabilities in network and application environments to assess their security posture.

Advanced Malware Analysis – Ransomware Forensics (2023)

- Reverse-engineered ransomware strains exploiting EternalBlue vulnerability on Windows.
- Performed timeline analysis, memory forensics, and Windows event log correlation to identify malicious behaviours.
- Demonstrated detection strategies and mitigation techniques for enterprise SOC environments.

HackAREA (2022)

- Led the development of "HackAREA," a beginner-friendly vulnerable web application designed to teach web security by exploiting and mitigating for OWASP Top 10 and SANS 25.

VOLUNTARY WORK

Secretary – Advanced Technology - *Warwick Student Union, University of Warwick*

WMG Volunteer Career Ambassador - *Warwick Careers Team, University of Warwick*

Lead Student Placement Representative - *Career Services Department – UPES*